

04_V2V_V2I_COMMUNICATION_RESULTS

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EVIDENCE PROVENANCE: [BENCH RF TELEMETRY]

Empirical measurement of dual-link 433 MHz LoRa and Wi-Fi transmission latency, airtime, and packet delivery rate.

1. Time Division Multiplexing (TDM) Schedule

To prevent packet collisions over half-duplex 433 MHz LoRa, a 2000 ms frame is partitioned: 0-500ms guard interval, 500-1000ms Truck 01 slot, 1000-1500ms Truck 02 slot, 1500-2000ms gateway broadcast. Wire format strictly preserves Rule 2: STATE,ID,seq,rpm,speed,ax,ay,az,gx,gy,gz.

2. Measured Communication Metrics

Communication Metric	Wi-Fi V2I Transport	LoRa V2V Transport	Safety Requirement
Frequency / Channel	2.412 GHz (Ch 1)	433.0 MHz (SF7/BW125)	Approved ISM band
Measured Airtime	1.2 ms	38.5 ms (SF7)	< 50.0 ms
End-to-End Transport Latency	12.4 ms avg	42.0 ms avg	< 250.0 ms
Packet Delivery Rate (PDR)	99.4%	98.7%	> 95.0%
Received Signal Strength (RSSI)	-48 dBm	-58 dBm (Bench)	> -85 dBm
Signal-to-Noise Ratio (SNR)	+24 dB	+9.5 dB	> -5.0 dB